1. How does a protective response work?

Page A: retrieve and practise. Use extra paper if needed. Keep your first attempt.

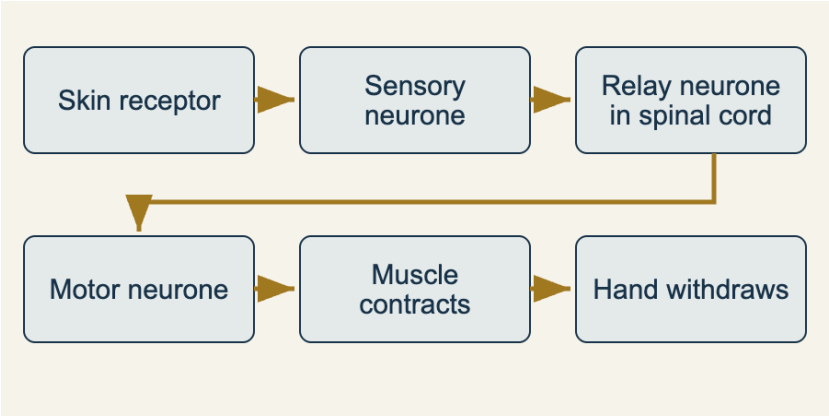
Candidate code: _____ Date: _____

Spot the change

Write two short answers in your practice record.

A hand touches a cold cup. What change can the skin detect?

Which part contracts to move the hand: a muscle or a bone?



Repair the safety diagram

Work with a partner. Each of you records a corrected route.

A trainee drew: muscle -> sensory neurone -> spinal cord -> motor neurone -> skin receptor.

Correct the order. Label where the stimulus is detected and where the response happens.

Check your understanding

How does a message pass through this nervous pathway?

A. A chemical messenger travels along the entire sensory neurone.

B. An electrical impulse travels along a neurone; a chemical messenger crosses a synapse.

C. An electrical impulse jumps across every synaptic gap.

1. Independent practice and retry

Page B: first attempt -> checked improvement -> a different unaided task.

Candidate code: _____ Date: _____

Make a safety induction card

Create a four-panel card for a new workshop trainee in your record.

Use the example of pulling a hand away from a sharp edge.

Show detection, the route through the CNS, transmission at one synapse and the muscle response.

Use arrows and short captions. Do not test pain or touch a sharp object.

Improve one link

Keep your first attempt. Open the review when asked and change one weak link.

Underline the sentence you changed. Tell your partner why the change improves the science.

A different protective response

Close the support. Answer independently in your record.

An eyelash touches the surface of an eye. A receptor detects it. Explain how a muscle can receive a message to close the eyelid.

Name the CNS region involved in this reflex.

2. Which messenger fits the job?

Page A: retrieve and practise. Use extra paper if needed. Keep your first attempt.

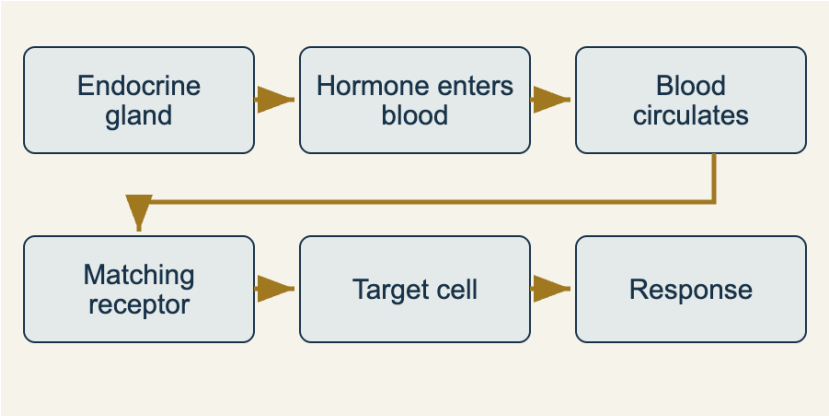
Candidate code: _____ Date: _____

Recover the route

Write the missing parts in your record.

Receptor -> _____ neurone -> CNS -> _____ neurone -> effector.

What crosses a chemical synapse?



Compare like with like

Complete the two gaps with a partner. Record your own table.

Feature | Nervous | Endocrine

Route | Neurones and synapses | _____

Usual speed | Very fast | Slower

Usual duration | _____ | Longer-lasting

Check your understanding

A hormone passes many cells but only some respond. Why?

A. Blood carries it only to those cells.

B. All cells respond in exactly the same way.

C. Only some cells have the appropriate receptors.

2. Independent practice and retry

Page B: first attempt -> checked improvement -> a different unaided task.

Candidate code: _____ Date: _____

Fix a technician's information card

Rewrite this card for a new laboratory technician in your record.

"Nerves send hormones through the blood. Hormones reach only their target organ. All hormonal responses are slow."

Make a short comparison table and add two sentences correcting the card. Use a named example.

Make the comparison useful

Use the review when asked. Improve one mismatched comparison or inaccurate claim.

Underline your change. Explain how it helps the new technician.

Same blood, different response

Close the support. Explain independently in your record.

Cells P and Q are exposed to the same hormone concentration. P responds; Q does not. Suggest a receptor-based explanation.

State one way this route differs from a motor neurone message.

3. How does the body correct a change?

Page A: retrieve and practise. Use extra paper if needed. Keep your first attempt.

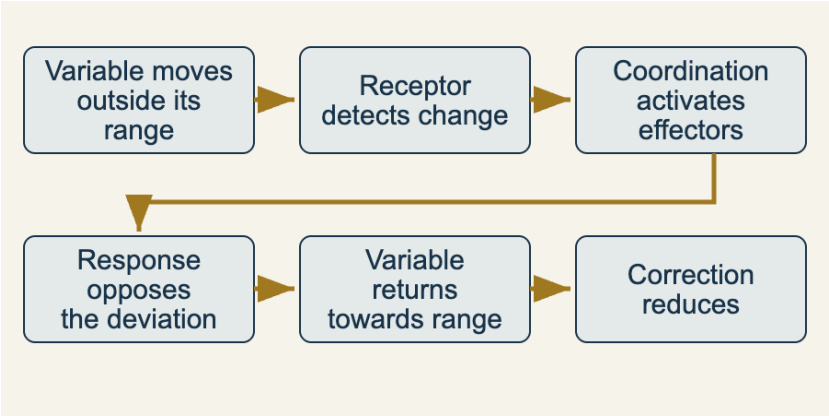
Candidate code: _____ Date: _____

Stable does not mean frozen

Record two short answers.

Why do working muscles need glucose?

Name a body condition that should stay within a suitable range.



Complete the opposite case

Complete the loop with a partner, then explain it in your own record.

Model incubator: temperature below range -> sensor -> controller -> heater _____ -> temperature _____.

Draw a second loop where the faulty response makes cooling worse. Label the fault.

Check your understanding

What makes feedback negative?

A. The response opposes the original deviation.

B. It always lowers the controlled variable.

C. It damages the organism.

3. Independent practice and retry

Page B: first attempt -> checked improvement -> a different unaided task.

Candidate code: _____ Date: _____

Repair the control plan

Work independently. Draw and explain a corrected two-way control plan.

A training model says: "If too high, raise it. If too low, lower it. Keep responding at full strength forever."

Replace all three rules. Use either a model incubator or a generic body variable.

Add one sentence explaining why a small fluctuation is not necessarily a failure.

Check the arrows, then improve

Compare your plan with the review. Correct one wrong causal link.

Mark your change in a different colour. Explain why the new arrow moves the variable towards its range.

Transfer the rule

Close the support. Use this new control rule independently.

In a model system, a dissolved substance drops below its chosen range. A controller reduces its removal from the liquid.

Explain why this could be negative feedback. State what should happen to the corrective response as the level recovers.

4. How does the body lose excess heat?

Page A: retrieve and practise. Use extra paper if needed. Keep your first attempt.

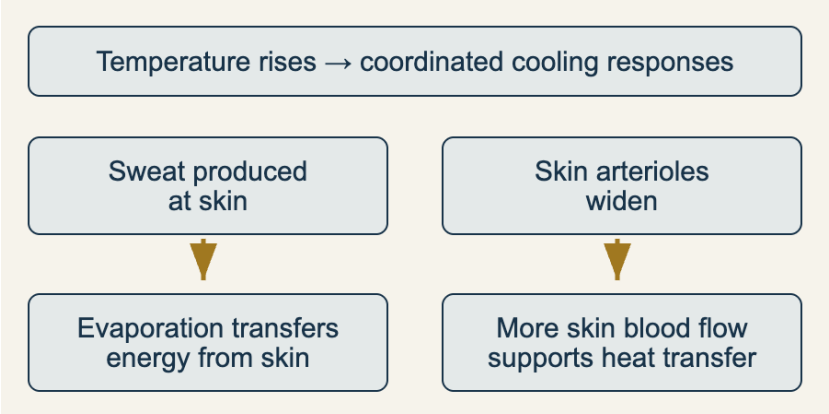
Candidate code: _____ Date: _____

Recover the control rule

Answer in your record.

An internal temperature rises above its suitable range. Should a corrective response increase or reduce heat loss?

Does negative feedback always make a variable fall?



Test two explanations

Work in pairs. Correct one sentence each, then record both.

"Sweat cools because cold water comes out."

"Blood vessels move up to the skin surface."

Check your understanding

In humid air, sweat evaporates less readily. What follows?

A. More visible sweat always means more cooling.

B. Cooling can be reduced even if the skin is sweaty.

C. Sweating never transfers energy.

4. Independent practice and retry

Page B: first attempt -> checked improvement -> a different unaided task.

Candidate code: _____ Date: _____

Create the cooling display

Create a three-part display in your record for the sports-session audience.

Explain sweating, skin blood flow and hair flattening with labelled arrows.

Add a note explaining why very humid air can make sweating less effective.

Keep the explanation scientific; follow the school's activity and heat procedures.

Improve a mechanism

Use the review to repair one arrow or caption. Keep the original visible.

Explain your change to the teacher.

Now the room is cold

Close the support. Answer independently in your record.

A resting person moves into a cold room. Explain how vasoconstriction and shivering help maintain core temperature.

5. How is blood glucose controlled?

Page A: retrieve and practise. Use extra paper if needed. Keep your first attempt.

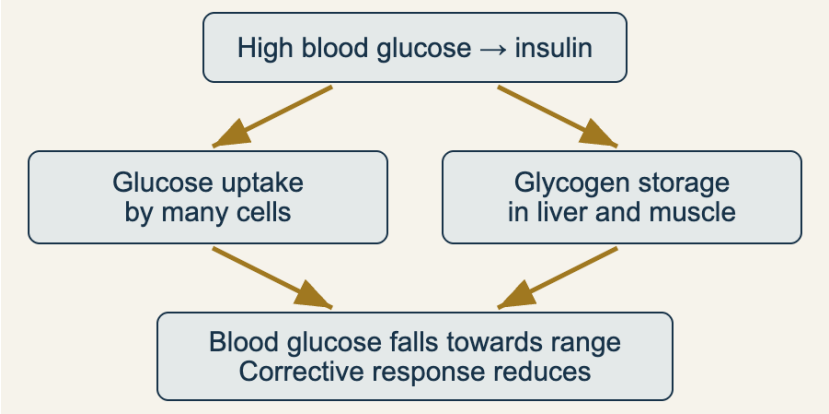
Candidate code: _____ Date: _____

Fuel and messages

Write two answers in your record.

Which process uses glucose to release usable energy?

How are hormones transported from a gland to target cells?



Sort three similar words

With a partner, match each word to its job. Record your own links.

Glucose: fuel molecule, storage carbohydrate or hormone?

Glycogen: fuel molecule, storage carbohydrate or hormone?

Glucagon: fuel molecule, storage carbohydrate or hormone?

Check your understanding

Blood glucose falls below its normal range. Which response helps restore it?

A. Insulin makes more glucose enter storage.

B. Glucagon changes into glycogen in the blood.

C. Glucagon promotes liver glucose release.

5. Independent practice and retry

Page B: first attempt -> checked improvement -> a different unaided task.

Candidate code: _____ Date: _____

Write a trainee's explanation

Work independently. Create two linked flow diagrams in your record.

Show the response to a glucose rise after a meal and the response to a low level between meals.

Compare the two meal profiles. Use one numerical difference and suggest why both levels later fall.

Use glucose, glycogen and glucagon accurately.

Synthetic comparison data (time/min: 0, 30, 60, 120): A = 5, 8, 6, 5; B = 5, 6, 6, 5 mmol L^-1.

Repair the confusing word

Use the review. Correct one word or causal arrow and explain the improvement.

Keep the first version visible.

Reject a different wrong claim

Close support. Correct this claim independently in your record.

"Insulin lowers blood glucose by changing into glucose. Glucagon stores glucose whenever the level is low."

6. What can the graphs really tell us?

Page A: retrieve and practise. Use extra paper if needed. Keep your first attempt.

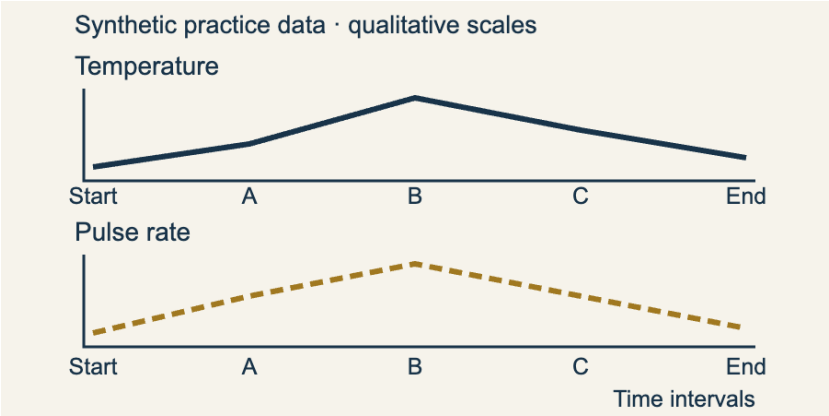
Candidate code: _____ Date: _____

Retrieve two mechanisms

Record two short answers.

What must sweat do to cool the skin effectively?

Which pancreatic hormone helps lower high blood glucose?



Repair the graph commentary

Correct the three claims with a partner, then record your own version.

"The peak happens at 30 minutes."

"The two lines meet, so the quantities are equal."

"The falling temperature means all heat production has stopped."

Check your understanding

Two variables rise together. What can the graph alone establish?

A. The first definitely causes the second.

B. They show an association in these data.

C. There is no useful pattern.

6. Independent practice and retry

Page B: first attempt -> checked improvement -> a different unaided task.

Candidate code: _____ Date: _____

Brief a new teacher

Write a short evidence note independently in your record.

Describe two linked trends in the practice graphs using the labelled intervals.

Suggest a cooling mechanism for the falling temperature.

State one thing the graphs cannot establish.

Make the evidence visible

Use the review. Improve one unsupported claim in your note.

Underline the graph evidence and keep the explanation separate.

An unlabeled time scale

Close the support. Answer independently in your record.

A new graph shows a concentration rising to a peak and then returning towards its starting level. Its time axis has no numerical labels.

Give one valid trend statement. Explain why "it recovers in exactly 10 minutes" is unjustified.

7. Return to cells: match structure to job

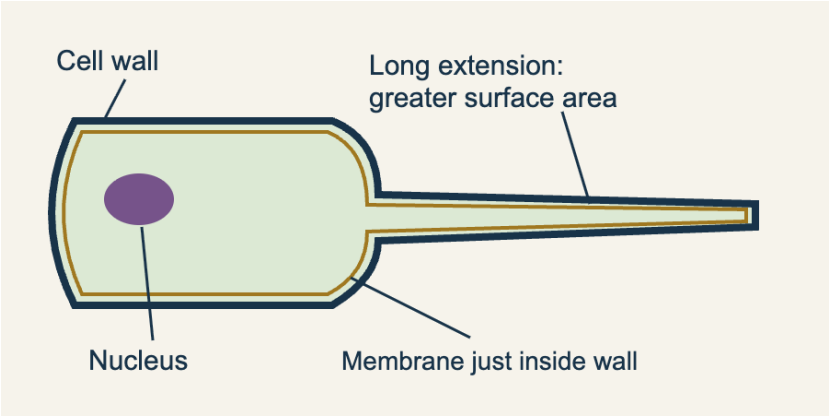
Page A: retrieve and practise. Use extra paper if needed. Keep your first attempt.

Candidate code: _____ Date: _____

Recover cell knowledge

Draw a simple plant cell in your record and label three structures.

Add one structure also found in an animal cell.



Complete a new caption

Work in pairs, then record your own caption.

A ciliated epithelial cell lines an airway. Its cilia beat to move mucus containing trapped particles.

Complete: "The cell has ___, which ___, helping ___."

Check your understanding

Which is a complete explanation?

A. The cell has a long extension.

B. The extension makes the cell photosynthesise.

C. The extension increases surface area for absorption.

7. Independent practice and retry

Page B: first attempt -> checked improvement -> a different unaided task.

Candidate code: _____ Date: _____

Create a cell-training card

Independently create a labelled root hair or ciliated cell card in your record.

Show three relevant structures. Add their functions.

Explain one specialised feature using a feature -> effect -> job chain.

Keep the two cell types distinct.

Turn a label into an explanation

Use the review. Improve one caption that only names a feature.

Show the change to the teacher before the fresh task.

A changed cell

Close support. Explain independently in your record.

A root hair cell develops a much shorter extension, with other features unchanged. Predict one effect on absorption and explain it.

8. Return to DNA: build an accurate model

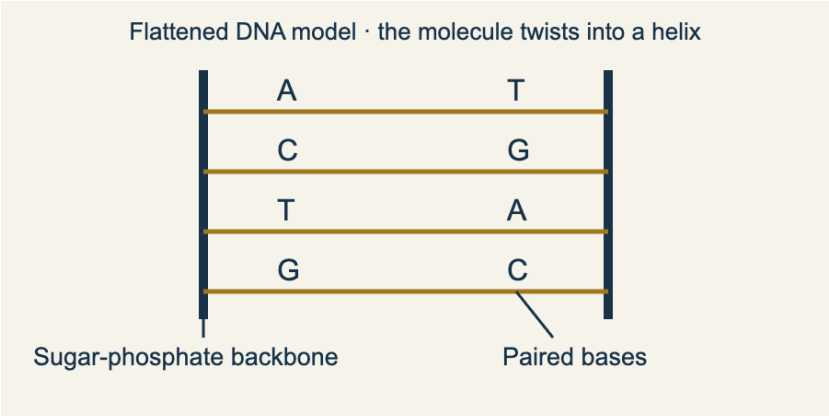
Page A: retrieve and practise. Use extra paper if needed. Keep your first attempt.

Candidate code: _____ Date: _____

Recall the extraction

Write what you observed and what you could not see.

Could you see individual base pairs in the extracted material with your unaided eyes?



Place the parts inside the whole

Draw the relationship with a partner, then label your own version.

A chromosome contains a long DNA molecule with many genes.

A gene is a section of DNA whose sequence provides information for a functional product, often a protein.

Proteins contribute to cell structures and processes, which can affect characteristics.

Check your understanding

What pairs with A G C in this simplified DNA model?

A. A G C

B. T C G

C. T G C

8. Independent practice and retry

Page B: first attempt -> checked improvement -> a different unaided task.

Candidate code: _____ Date: _____

Build a training model

Create an annotated model in your record using the strand G T A C C.

Add the complementary bases and label backbone, base pair and double helix.

Write a caption explaining a gene and one limitation of your model.

Correct the representation

Use the review. Correct a mismatched pair or unclear gene caption.

Keep the original and explain the correction.

A fresh sequence and a claim

Close support. Complete the sequence and correct the claim independently.

Find the complementary strand to C A G T.

Correct: "The white threads proved we could see individual genes."

9. Return to inheritance: reason from a family

Page A: retrieve and practise. Use extra paper if needed. Keep your first attempt.

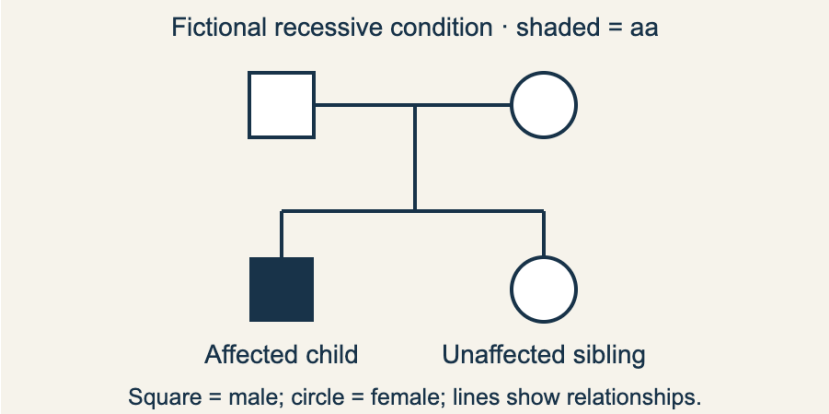
Candidate code: _____ Date: _____

Recover the allele rules

Record two answers using the stated classroom model.

R is dominant for a fictional seed pattern; r is recessive. Which phenotype does Rr show?

Which allele can an rr parent pass on?



Complete the family reasoning

Record a cross for the two parents, then discuss the possible unaffected genotypes.

Use Aa x Aa for this practice pedigree. Label the gametes before filling the boxes.

Check your understanding

In the recessive practice pedigree, what genotype can an unaffected sibling have?

A. Only AA.

B. Only Aa.

C. AA or Aa.

9. Independent practice and retry

Page B: first attempt -> checked improvement -> a different unaided task.

Candidate code: _____ Date: _____

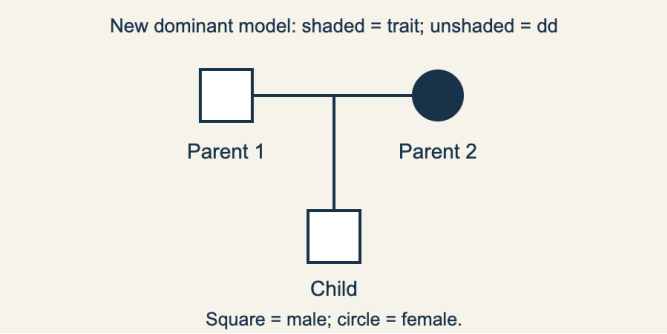
Explain a new family model

Work independently in your record. State the model before answering.

Use this new pedigree: shaded means a fictional dominant trait (D); unshaded means no trait (dd). Infer the genotype of each parent from their child.

Draw the parental cross and explain each phenotype's probability.

A different person shows the dominant trait, with no family evidence. State all possible genotypes.



Remove the unsupported certainty

Use the review. Correct one genotype claim or probability statement.

Keep your first response and explain your change to the teacher.

New letters, same reasoning

Close support. Write your answers independently using the stated fictional model.

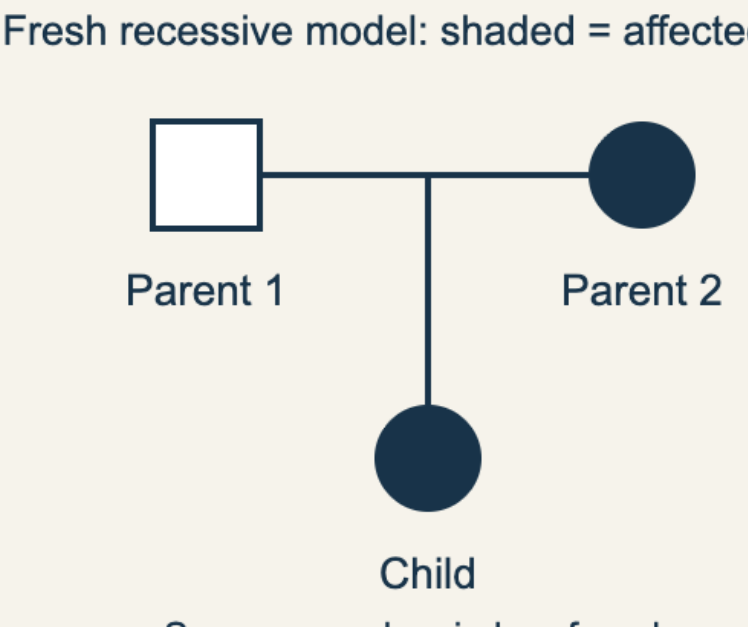
In a new fictional recessive family, shaded means affected (bb). Use the pedigree to infer both parents' genotypes and justify each inference.

Give the probability that their next child is affected. Does the first child change that probability?

9. Fresh pedigree for the unaided retry

Use this diagram with the retry question on the previous page.

Candidate code: _____ Date: _____



New letters, same reasoning

Close support. Write your answers independently using the stated fictional model.

In a new fictional recessive family, shaded means affected (bb). Use the pedigree to infer both parents' genotypes and justify each inference.

Give the probability that their next child is affected. Does the first child change that probability?